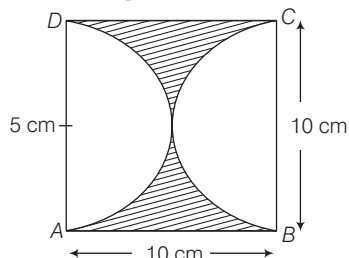


67. (a) Let $b = 4x$ and $l = 9x$
 $\therefore$ Area of rectangle $= l \times b$
 $144 = 4x \times 9x$
 $\Rightarrow x^2 = \frac{144}{36} \Rightarrow x^2 = 4 \therefore x = 2$
 Now, $b = 4 \times 2 = 8$ cm and
 $l = 9 \times 2 = 18$ cm
 $\therefore$ Perimeter of rectangle $= 2(b + l)$
 $= 2(8 + 18) = 2 \times 26 = 52$ cm

68. (d) Area between square and semi-circles
 $=$ Area of square $- 2$ (Area of semi-circle)



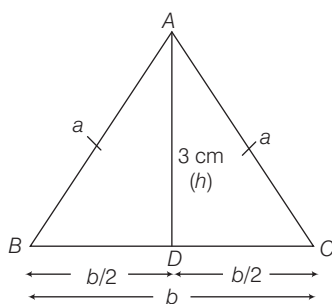
$$= (10)^2 - 2 \times \frac{1}{2} \times \frac{22}{7} \times (5)^2 = 100 - 78.5$$

$$= 21.5 \text{ cm}^2$$

Hence, the area between the square and semi-circular cross-section is 21.5 cm^2 .

69. (a) $1 \text{ m} = 1000 \text{ mm}$
 $\therefore 10 \text{ m} = 10000 \text{ mm}$
 Number of 200 mm lengths that can be cut from 10 m of ribbon
 $= \frac{10000}{200} = 50$

70. (a) Area of the $\triangle ABC = \frac{1}{2} \times b \times h$



$$\Rightarrow 12 = \frac{1}{2} \times b \times 3$$

$$\therefore b = \frac{12 \times 2}{3} = 8 \text{ cm}$$

$$\text{Here, } BD = CD = \frac{b}{2} = \frac{8}{2} = 4 \text{ cm}$$

In right angled $\triangle ABD$, by pythagoras theorem,

$$AB = \sqrt{BD^2 + AD^2}$$

$$\Rightarrow a = \sqrt{4^2 + 3^2} = \sqrt{16 + 9}$$

$$= \sqrt{25} = 5 \text{ cm}$$

Now, perimeter of an isosceles triangle

$$= 2a + b = 2 \times 5 + 8$$

$$= 10 + 8 = 18 \text{ cm}$$

71. (a) Area of equilateral triangle

$$= \frac{\sqrt{3}a^2}{4} = x \quad \dots(i)$$

$$\text{and perimeter} = 3a = y \Rightarrow a = \frac{y}{3} \dots(ii)$$

Now, putting the value of a from Eq. (ii) in Eq. (i), we get

$$\frac{\sqrt{3} \left(\frac{y}{3} \right)^2}{4} = x \Rightarrow x = \frac{\sqrt{3} \times y^2}{9 \times 4}$$

$$\Rightarrow x = \frac{y^2}{3\sqrt{3} \times 4} \Rightarrow x = \frac{y^2}{12\sqrt{3}}$$

$$\Rightarrow 12\sqrt{3} x = y^2$$

On squaring both sides, we get

$$y^4 = 432x^2$$

72. (b) Here, $a = 7$ cm, $b = 24$ cm

$$\text{and } c = 25 \text{ cm}$$

Semi-perimeter of triangle

$$s = \frac{a + b + c}{2} = \frac{7 + 24 + 25}{2} = \frac{56}{2}$$

$$= 28 \text{ cm}$$

According to the question,

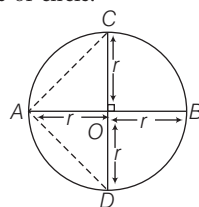
Area of circle = Area of triangle

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{28(28-7)(28-24)(28-25)}$$

$$= \sqrt{28 \times 21 \times 4 \times 3} = \sqrt{7056} = 84 \text{ cm}^2$$

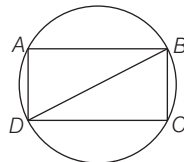
73. (b) Diameters of a circle intersect at the centre of circle.



$$\text{Required ratio} = \frac{\text{Area of circle}}{\text{Area of } \triangle ACD}$$

$$= \frac{\pi r^2}{\frac{1}{2} \times 2r \times r} = \pi$$

74. (c) Area of rectangle inscribed in a circle is maximum, when



Diameter of the circle = Diagonal of the rectangle

Now, let x and y be the length and breadth of the rectangle, respectively.

Now in $\triangle ABD$, by pythagoras theorem

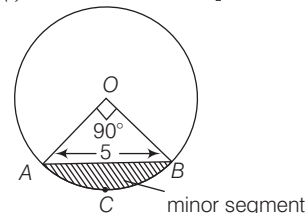
$$AB^2 + AD^2 = (5)^2 \Rightarrow x^2 + y^2 = 25$$

Since, they form pythagorean triplet,

$$\therefore x = 4 \text{ and } y = 3$$

$$\text{So, area of rectangle} = 3 \times 4 = 12 \text{ cm}^2$$

75. (c) Let $AO = OB = r$ [radius of circle]



In $\triangle AOB$, by pythagoras theorem,

$$AB^2 = OA^2 + OB^2$$

$$\Rightarrow (5)^2 = r^2 + r^2 \quad [\because AB = 5 \text{ cm}]$$

$$\therefore r^2 = \frac{25}{2} \text{ cm}$$

Now, area of sector

$$ACBOA = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \frac{90^\circ}{360^\circ} \times \pi \times \frac{25}{2} = \frac{25\pi}{8} \text{ cm}^2$$

Now, area of minor segment $ACBA$

= Area of sector $ACBOA$ - Area of triangle

$$= \frac{25\pi}{8} - \frac{r^2}{2} = \frac{25\pi}{8} - \frac{25}{4}$$

$$= \left(\frac{25\pi - 50}{8} \right) \text{ cm}^2$$

Area of major segment = Area of circle - Area of minor segment

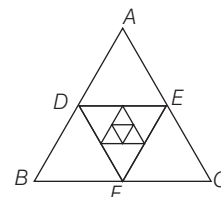
$$= \pi r^2 - \left(\frac{25\pi - 50}{8} \right)$$

$$= \frac{25\pi}{2} - \frac{(25\pi - 50)}{8}$$

$$= \frac{100\pi - 25\pi + 50}{8} = \frac{75\pi + 50}{8}$$

$$= \frac{25}{8} (3\pi + 2) = \frac{25}{4} \left(\frac{3\pi}{2} + 1 \right) \text{ cm}^2$$

76. (c)



Perimeters of triangles

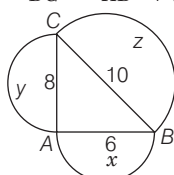
$$= 3 + 3 \times \frac{1}{2} + 3 \times \frac{1}{4} + 3 \times \frac{1}{8} + \dots$$

$$= 3 \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right] = 3 \times \frac{1}{1 - \frac{1}{2}}$$

$$= 6 \text{ units}$$

77. (c) In $\triangle ABC$, by pythagoras theorem,

$$BC^2 = AB^2 + AC^2$$



$$= 6^2 + 8^2 = 36 + 64 = 100 \text{ cm}$$

$$\therefore BC = 10 \text{ cm}$$

$$\text{Radius of semi-circle } x = \frac{6}{2} = 3$$

$$\text{Now, area of semi-circle} = x = \frac{\pi(3)^2}{2} = \frac{9\pi}{2} \text{ cm}^2$$

$$\text{Radius of semi-circle, } y = \frac{8}{2} \text{ cm}$$

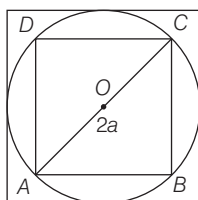
$$\text{Area of semi-circle} = y = \frac{16\pi}{2} \text{ cm}^2$$

$$\text{Radius of semi-circular, } z = \frac{10}{2} = 5 \text{ cm}^2$$

$$\text{Area of semi-circle} = z = \frac{25\pi}{2} \text{ cm}^2$$

$$\text{Now, } x + y - z = \frac{9\pi}{2} + \frac{16\pi}{2} - \frac{25\pi}{2} = 0$$

78. (b)



For inscribed circle, Diameter of circle = Diagonal of square = $2a$

Now, in $\triangle ABC$ by pythagoras theorem,

$$AB^2 + BC^2 = AC^2 \Rightarrow 2AB^2 = 4a^2$$

$$AB^2 = 2a^2 \Rightarrow AB = \sqrt{2}a$$

$\therefore$ Area of inner square

$$= AB^2 = (\sqrt{2}a)^2 = 2a^2$$

For circumscribed square,

Diameter of a circle = Side of square

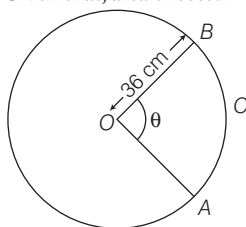
$\therefore$ Area of circumscribed square

$$= (2a)^2 = 4a^2$$

$\therefore$ Difference between areas of outer and inner squares

$$= 4a^2 - 2a^2 = 2a^2$$

79. (d) Given that, area of sector = $72\pi \text{ cm}^2$



$$\Rightarrow \frac{\pi r^2 \theta}{360^\circ} = 72\pi$$

$$\Rightarrow \theta = \frac{72 \times 360}{36 \times 36} = 20^\circ$$

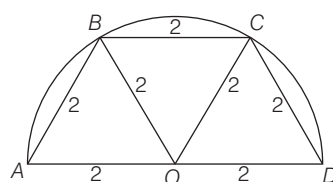
$$\begin{aligned} \text{Now, length of arc } ACB &= \frac{\pi r \theta}{180^\circ} \\ &= \frac{\pi \times 36 \times 20}{180} = 4\pi \text{ cm} \end{aligned}$$

80. (d) Since, $\triangle AOB$, $\triangle BOC$ and $\triangle COD$ are equilateral.

$$\therefore \text{Side of an equilateral triangle} = \text{radius of semi-circle} = 2 \text{ cm}$$

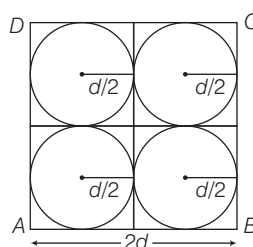
Now, total area of three equilateral triangles

$$= 3 \times \frac{\sqrt{3}}{4} (\text{Side})^2$$



$$= 3 \times \frac{\sqrt{3}}{4} \times 4 = 3\sqrt{3} \text{ cm}^2$$

81. (c)



From the figure it is clear that, 4 circular plates of diameter d can be made from.

square plate of side $2d$ with minimum loss of material.

82. (d) Given, sides of a triangle are in the ratio = $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$ i.e. $6 : 4 : 3$

Let the sides of a triangle be $6x$, $4x$ and $3x$, respectively.

We know that,

Perimeter of a triangle = Sum of all sides of a triangle

$$\Rightarrow 52 = 6x + 4x + 3x$$

$$\Rightarrow 52 = 13x$$

$$\therefore x = \frac{52}{13} = 4$$

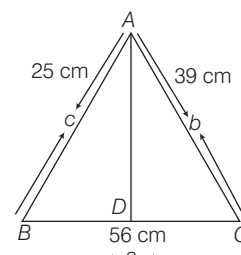
$\therefore$ Smallest side of a triangle

$$= 3x = 3 \times 4 = 12 \text{ cm}$$

83. (c) Let the three sides of a triangle be a , b and c , respectively.

$$\therefore a = 56 \text{ cm, } b = 39 \text{ cm}$$

$$\text{and } c = 25 \text{ cm}$$



Perimeter of triangle, $2s = a + b + c$

$$\Rightarrow 2s = 25 + 39 + 56$$

$$\Rightarrow s = 60 \text{ cm}$$

Now, area of triangle

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{60(60-25)(60-39)(60-56)}$$

$$= \sqrt{60 \times 35 \times 21 \times 4} = 420 \text{ cm}^2 \quad \dots(i)$$

$$\text{Again area of triangle} = \frac{1}{2} \times \text{Base}$$

$\times$ Altitude

$$= \frac{1}{2} \times BC \times AD \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{1}{2} \times BC \times AD = 420$$

$$\Rightarrow \frac{1}{2} \times 56 \times AD = 420$$

$$\Rightarrow AD = \frac{420}{28} = 15 \text{ cm}$$

Hence, the length of AD is 15 cm.

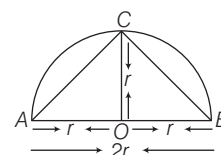
84. (a) Let the r be the radius of semi-circle.

In $\triangle ABC$,

$$AO = OB = OC = r \text{ [radius of circle]}$$

$\therefore$ Area of largest triangle

$$= \frac{1}{2} \times \text{Base} \times \text{Height}$$



$$= \frac{1}{2} \times AB \times OC = \frac{1}{2} \times 2r \times r$$

$$[\because AB = OA + OB]$$

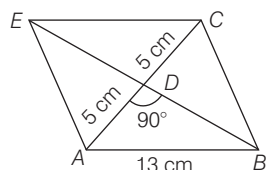
$$= r^2$$

Hence, the area of the largest triangle

is r^2 .

85. (c) Given, a side of rhombus (a) = 13 cm and a diagonal of rhombus (d_1) = 10 cm

Now, in right angled $\triangle ABD$,



$$\begin{aligned} AB^2 &= AD^2 + BD^2 \\ \Rightarrow 13^2 &= 5^2 + BD^2 \\ \Rightarrow BD^2 &= 169 - 25 = 144 \\ \therefore BD &= 12 \text{ cm} \\ \therefore EB &= 2 \times BD = 2 \times 12 = 24 \text{ cm} \\ \text{Length of second diagonal } (d_2) &= 24 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Now, area of rhombus} &= \frac{1}{2} \times d_1 \times d_2 \\ &= \frac{1}{2} \times 10 \times 24 = 120 \text{ cm}^2 \end{aligned}$$

Shortcut Method

Here, $a = 13$ cm, $d_1 = 10$ cm

We know that,

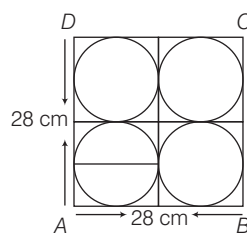
$$\begin{aligned} 4a^2 &= d_1^2 + d_2^2 \\ \Rightarrow 4 \times 13^2 &= 10^2 + d_2^2 \\ \Rightarrow 4 \times 169 &= 100 + d_2^2 \\ \Rightarrow 676 - 100 &= d_2^2 \\ \Rightarrow d_2 &= \sqrt{576} = 24 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Now, area of rhombus} &= \frac{1}{2} d_1 d_2 \\ &= \frac{1}{2} \times 10 \times 24 = 120 \text{ cm}^2 \end{aligned}$$

86. (d) Let the side of a square paper sheet be a cm.

Given, area of square paper sheet = 784 cm²

$$\Rightarrow a^2 = 784 \Rightarrow a = 28 \text{ cm}$$



$$\therefore \text{Diameter of one circle} = \frac{28}{2} = 14 \text{ cm}$$

$$\text{and radius of one circle} = \frac{14}{2} = 7 \text{ cm}$$

$$\begin{aligned} \therefore \text{Circumference of each plate} &= 2\pi r \\ &= 2\pi \times 7 = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm} \end{aligned}$$

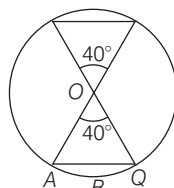
Hence, the circumference of each plate is 44 cm.

87. (b) Given, radius of circle, $r = 3$ cm

Area of remaining portion

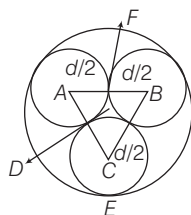
= Area of circle - 2 (Area of sector OABQO).

$$\begin{aligned} &= \pi r^2 - \frac{2\pi r^2 \theta}{360^\circ} \\ &= \pi r^2 \left[1 - \frac{2\theta}{360^\circ} \right] \\ &= \pi r^2 \left[1 - \frac{2 \times 40}{360^\circ} \right] \\ &= \frac{22}{7} \times 9 \left[1 - \frac{8}{36} \right] \\ &= \frac{22}{7} \times 9 \left[1 - \frac{2}{9} \right] = \frac{22}{7} \times \frac{7}{9} \times 9 = 22 \text{ cm}^2 \end{aligned}$$



88. (d) Given, diameter of each circle = d

$\therefore$ Radius of each circle = $d/2$



$$CE = AF = BF = d/2$$

Here, $\triangle ABC$ is an equilateral triangle.

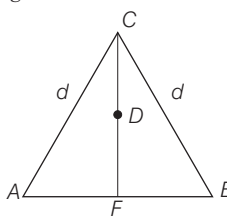
$\therefore$ Altitude of equilateral triangle

$$= \frac{\sqrt{3}}{2} (\text{Side}) = \frac{\sqrt{3}}{2} (AB) = \frac{\sqrt{3}}{2} d$$

$$[\because AB = AF + FB = d]$$

Let D be the centroid of $\triangle ABC$

$\therefore$ Ratio of centroid of equilateral triangle = 2 : 1



$$\therefore \text{Length of } DC = \frac{2}{3} \times \frac{\sqrt{3}}{2} d = \frac{\sqrt{3}}{3} d$$

Now, radius of big circle,

$$\begin{aligned} R = DE &= DC + CE = \frac{\sqrt{3}}{3} d + \frac{d}{2} \\ &= d \left[\frac{\sqrt{3}}{3} + \frac{1}{2} \right] = \frac{d \times \sqrt{3}(2 + \sqrt{3})}{6} \end{aligned}$$

$\therefore$ Area of big circle = πR^2

$$\begin{aligned} &= \pi \left[d \times \frac{\sqrt{3}(2 + \sqrt{3})}{6} \right]^2 \\ &= \pi \times \left[\frac{d^2 \times 3(2 + \sqrt{3})^2}{36} \right] \end{aligned}$$

$$= \frac{\pi d^2 (2 + \sqrt{3})^2}{12}$$

Hence, the area of big circle is

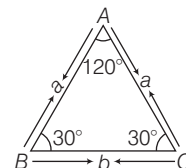
$$= \frac{\pi d^2 (2 + \sqrt{3})^2}{12}$$

89. (c) Let the angles of a triangle be $4x$, x and x , respectively.

$$\begin{aligned} \text{i.e. } \angle A &= 4x, \\ \angle B &= x \quad \text{and} \\ \angle C &= x \end{aligned}$$

$\therefore$ Sum of all angles of a triangle = 180°

$$\therefore \angle A + \angle B + \angle C = 180^\circ$$



$$\Rightarrow 4x + x + x = 180^\circ \Rightarrow 6x = 180^\circ$$

$$\Rightarrow x = 30^\circ$$

$$\therefore \angle A = 4x = 30 \times 4 = 120^\circ,$$

$$\angle B = x = 30^\circ$$

$$\text{and } \angle C = x = 30^\circ$$

So, it is clear that, given triangle is isosceles triangle. Let the sides isosceles triangle be a , a and b , respectively.

$\therefore$ Perimeter of triangle

$$= a + a + b = 2a + b$$

$$\text{Using sine rule, } \frac{b}{\sin 120^\circ} = \frac{a}{\sin 30^\circ}$$

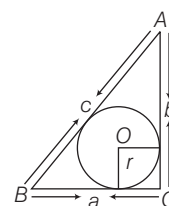
$$\Rightarrow \frac{b}{\sqrt{3}/2} = \frac{a}{1/2} \Rightarrow \frac{2b}{\sqrt{3}} = 2a \Rightarrow a = \frac{b}{\sqrt{3}}$$

$$\text{Now, } \frac{b}{2a + b} = \frac{b}{2 \times \frac{b}{\sqrt{3}} + b}$$

$$= \frac{b}{\frac{2b + \sqrt{3}b}{\sqrt{3}}} = \frac{\sqrt{3}}{2 + \sqrt{3}}$$

90. (a) Given, a , b and c be the sides of a right triangle, where c is the hypotenuse.

$$\therefore c^2 = a^2 + b^2 \quad \dots(i)$$



and perimeter of triangle $2s = a + b + c$

$$\therefore s = \frac{a + b + c}{2}$$

$$\text{Area of triangle} = \frac{1}{2} \times BC \times AC$$

$$\Rightarrow \Delta = \frac{1}{2} ab \quad \dots(ii)$$

From Eq. (i), $c^2 = a^2 + b^2$

$$\Rightarrow c^2 = (a + b)^2 - 2ab$$

$$\Rightarrow 2ab = (a+b)^2 - c^2$$

$$\Rightarrow ab = \frac{(a+b)^2 - c^2}{2}$$

Now, area of triangle = $\frac{ab}{2}$ [from Eq. (ii)]

$$\Delta = \frac{[(a+b)^2 - c^2]}{2}, \Delta = \frac{[(a+b)^2 - c^2]}{4}$$

$$\therefore \text{Radius of circle, } r = \frac{\Delta}{\frac{1}{2}(a+b+c)} = \frac{(a+b-c)}{4 \times \left(\frac{a+b+c}{2}\right)} = \frac{(a+b-c)}{2}$$

Hence, the radius of circle is $\frac{(a+b-c)}{2}$

91. (c) Let the radius of outer circle and inner circle be R and r , respectively.

Perimeter of outer circle = $2\pi R$

and perimeter of inner circle = $2\pi r$

$$\therefore \frac{2\pi R}{2\pi r} = \frac{23}{22}$$

$$\Rightarrow \frac{R}{r} = \frac{23}{22}$$

From the figure, $R = 5 + r$

$$\therefore \frac{5+r}{r} = \frac{23}{22} \Rightarrow 110 + 22r = 23r$$

$$\Rightarrow r = 110 \text{ m}$$

So, diameter of the inner circle is 220 m.

92. (a) Let the sides of a triangular field be a , b and c , respectively.

$$\therefore a = 41 \text{ m}, b = 40 \text{ m and } c = 9 \text{ m}$$

Perimeter of triangular field = $a + b + c$

$$\Rightarrow 2s = a + b + c = (41 + 40 + 9)$$

$$\Rightarrow 2s = 90 \Rightarrow s = 45 \text{ m}$$

$\therefore$ Area of triangular field

$$= \frac{\sqrt{s(s-a)(s-b)(s-c)}}{2}$$

$$= \frac{\sqrt{45(45-41)(45-40)(45-9)}}{2}$$

$$= \frac{\sqrt{45 \times 4 \times 5 \times 36}}{2}$$

$$= \frac{\sqrt{9 \times 5 \times 5 \times 4 \times 9 \times 4}}{2}$$

$$= 9 \times 5 \times 4 = 180 \text{ m}^2$$

Given, area of each rose bed = 900 cm^2

$$= \frac{900}{100 \times 100} \text{ m}^2$$

$\therefore$ Number of rose beds

$$= \frac{\text{Area of triangular field}}{\text{Area of one rose bed}}$$

$$= \frac{180 \times 100 \times 100}{900} = 2000$$

93. (b) 1 km = 1000 m 1 dm = 10 m

$$\therefore 2 \text{ km and } 26 \text{ dm}$$

$$= 2000 + 260 = 2260 \text{ m}$$

Let radius be r m.

Distance covered in revolution = $2\pi r$

$$\text{Distance covered in 452 revolution}$$

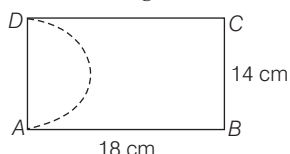
$$= 452 \times 2\pi r$$

$$\therefore 452 \times 2\pi r = 2260$$

$$r = \frac{2260 \times 7}{452 \times 2 \times 22} = \frac{35}{44}$$

$$\text{Diameter} = 2 \times \frac{35}{44} = \frac{35}{22} = 1\frac{13}{22} \text{ m}$$

94. (d) Area of remaining portion = Area of rectangle - Area of semi-circle



$$= 18 \times 14 - \frac{1}{2} \times \frac{22}{7} \times 7 \times 7$$

$$= 252 - 77 = 175 \text{ cm}^2$$

95. (d) Let length of side of square and equilateral triangle be x and y , respectively.

$$\text{Then, } 4x = 3y \quad \dots(i)$$

Given, diagonal of square = $12\sqrt{2}$

$$\Rightarrow x\sqrt{2} = 12\sqrt{2}$$

$$\therefore x = 12 \text{ cm}$$

$$\therefore y = \frac{4x}{3} = \frac{4}{3} \times 12$$

$$= 16 \text{ cm}$$

$$\text{So, area of triangle} = \frac{\sqrt{3}}{4} \times 16 \times 16$$

$$= 64\sqrt{3} \text{ cm}^2$$

96. (d) Let the side of square be x .

Area of square = x^2

Now, increased length

$$= x \left(1 + \frac{30}{100} \right) = \frac{13x}{10}$$

and increased breadth

$$= x \left(1 + \frac{20}{100} \right) = \frac{12x}{10}$$

Now, new area of rectangle

$$= \frac{13x}{10} \times \frac{12x}{10} = \frac{156x^2}{100}$$

Increased in area

$$= \frac{156x^2}{100} - x^2 = \frac{56x^2}{100}$$

Percentage increased in area

$$= \frac{56x^2}{100x^2} \times 100$$

$$= 56\%$$

Shortcut Method

Here, $x = 30\%$, $y = 20\%$

$$\therefore \text{Net effect} = \left[x + y + \frac{xy}{100} \right] \%$$

$$= \left[30 + 20 + \frac{30 \times 20}{100} \right] = 56\%$$

Hence, area is increased by 56%.

97. (a) Area of trapezium

$$= \frac{1}{2} (\text{Sum of parallel side}) \times \text{Distance}$$

between them

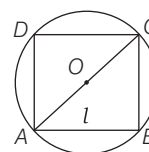
$$\Rightarrow \frac{1}{2} (5x + 7x) \times 14 = 336$$

$$\Rightarrow 12x \times 7 = 336 \Rightarrow x = 4$$

The length of smaller side

$$= 4 \times 5 = 20 \text{ cm}$$

98. (c) We have, $2\pi r = 100$



$$\therefore 2\pi r = \frac{100}{\pi} \text{ cm}$$

Let the side of square be l cm.

$$\therefore l\sqrt{2} = \frac{100}{\pi} \Rightarrow l = \frac{50\sqrt{2}}{\pi} \text{ cm}$$

99. (b) There are 437 fruits plants.

$$\therefore 437 = 19 \times 23$$

$\therefore$ There are 19 rows and 23 trees or there are 23 rows and 19 trees.

Given, the distance between any two adjacent plants is 2 m and the distance between any two adjacent rows is 2 m.

$\therefore$ Length of orchard

$$= [1 + 22(2) + 1] = 46 \text{ m}$$

Breadth of orchard

$$= [1 + 18(2) + 1] = 38 \text{ m}$$

$\therefore$ Perimeter of orchard

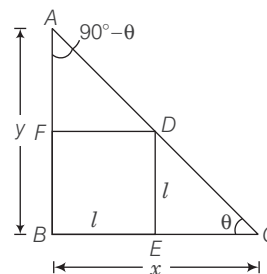
$$= [(46 \times 2) + (38 \times 2)] = 168 \text{ m}$$

$\therefore$ Cost of fencing at the rate of

$$= ₹ 100 \text{ per metre}$$

$$= 168 \times 100 = ₹ 16800$$

100. (b) Let length of the side of square be l .



$$\text{In } \triangle DEC, \tan \theta = \frac{DE}{EC} = \frac{l}{x-l} \dots(i)$$

$$\text{In } \triangle AFD, \tan (90^\circ - \theta) = \cot \theta = \frac{FD}{AF}$$

$$= \frac{l}{y-l} \dots(ii)$$

On multiplying Eqs. (i) and (ii), we get

$$1 = \frac{l^2}{(x-l)(y-l)}$$

$$\Rightarrow xy - xl - yl + l^2 = l^2$$

$$\therefore l(x+y) = xy \Rightarrow l = \frac{xy}{x+y}$$

$$\therefore \text{Perimeter of square} = 4l = \frac{4xy}{x+y}$$

101. (c) Let a be the side of regular polygon then, inradius, $r = \frac{a}{2 \tan\left(\frac{\pi}{n}\right)}$

$$\Rightarrow a = 2r \tan\left(\frac{\pi}{n}\right)$$

$\therefore$ Perimeter of polygon,

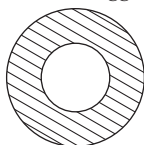
$$p = na = 2nr \tan\left(\frac{\pi}{n}\right)$$

$$\Rightarrow \tan\left(\frac{\pi}{n}\right) = \frac{p}{2nr} \quad \dots(i)$$

$$\text{Area of polygon} = nr^2 \tan\left(\frac{\pi}{n}\right)$$

$$= nr^2 \times \frac{p}{2nr} = \frac{pr}{2} \quad [\text{from Eq. (i)}]$$

102. (c) Let the radius of smaller ring be r_1 and the radius of bigger ring be r .



Since, the smaller ring takes three full revolutions to roll over the circumference of bigger ring.

$$\therefore 2\pi r = 3(2\pi r_1) \Rightarrow r = 3r_1$$

$$\text{Area of pathway} = \pi r^2 - \pi r_1^2 = n(\pi r_1^2)$$

$$\Rightarrow \pi r^2 = (n+1)\pi r_1^2$$

$$\Rightarrow r^2 = (n+1)r_1^2 \Rightarrow n = 9 - 1 = 8$$

103. (d) Number of rounds that a wheel of diameter D will make in traversing a distance $= n(\pi D)$

Given that, distance $= 4 \text{ km}$

$$\Rightarrow n(\pi D) = 4000 \text{ m}$$

$$\Rightarrow n\left(\pi \times \frac{7}{11}\right) = 4000$$

$$\Rightarrow n\left(\frac{22}{7} \times \frac{7}{11}\right) = 4000$$

$$\Rightarrow n = 2000$$

Hence, the number of rounds is 2000.

104. (c) Here, base $= 300$ units and side $= 170$ units

We know that,

Area of an isosceles triangle

$$= \frac{\text{Base}}{4} \sqrt{4(\text{Side})^2 - (\text{Base})^2}$$

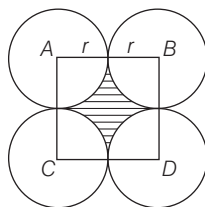
$$= \frac{300}{4} \sqrt{4(170)^2 - (300)^2}$$

$$= 75 \sqrt{4 \times 28900 - 90000}$$

$$= 75 \sqrt{115600 - 90000} = 75 \sqrt{25600}$$

$$= 75 \times 160 = 12000 \text{ sq units}$$

105. (d) Let the radius of the circle be r . Then, side of square $ABCD = 2r$



$\therefore$ Area of empty space $=$ Area of square $- 4$ [Area of quadrant]

$$\Rightarrow \frac{150}{847} = (2r)^2 - 4 \left[\frac{1}{4} \times \pi r^2 \right]$$

$$\Rightarrow \frac{150}{847} = \frac{6}{7} r^2$$

$$\Rightarrow r^2 = \frac{25}{121} \quad \left[\because \pi = \frac{22}{7} \right]$$

$$\Rightarrow r = \frac{5}{11} \text{ cm}$$

106. (c) Area of circle with diameter (AD)

$$= \frac{\pi(AD)^2}{4}$$

$$\text{So, } \frac{\pi(AD)^2}{4} = 707$$

$$\Rightarrow AD = \sqrt{\frac{707}{\pi}} \times 4 = 2\sqrt{\frac{707}{\pi}} \text{ m}$$

$$\text{As, } AB = BC = CD = \frac{AD}{3} = \frac{2}{3} \sqrt{\frac{707}{\pi}} \text{ m}$$

and $AC = 2AB = BD$

$$= 2CD = \frac{4}{3} \sqrt{\frac{707}{\pi}} \text{ m}$$

$\therefore$ Area of shaded portion $= 2$

[Area of semi-circle with AC as diameter $-$ Area of semi-circle with AB as diameter]

$$= \frac{2\pi(AC)^2}{8} - \frac{2\pi(AB)^2}{8}$$

$$= \frac{2\pi}{8} \left[\frac{16}{9} \cdot \frac{707}{\pi} - \frac{4}{9} \cdot \frac{707}{\pi} \right] = \frac{1}{3} \times 707 \text{ m}^2$$

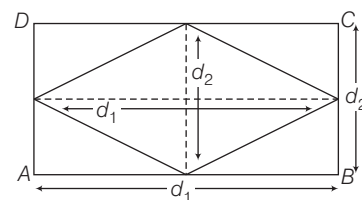
$\therefore$ Cost of levelling the shaded portion

$$= \frac{707}{3} \times 63 = ₹ 14847$$

107. (b) We know that, Area of rhombus

$$= \frac{1}{2} \times \text{product of diagonals}$$

$$= \frac{1}{2} \times d_1 \times d_2$$



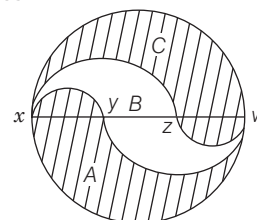
From the above figure,

$$\text{Area of rectangle} = d_1 \times d_2$$

$$= 2 (\text{Area of rhombus})$$

$$= 2 \times 2 = 4 \text{ sq units}$$

108. (b) In the given question, A radii of 3 m is divided in such a way that the radii of smaller semi-circle is 1 m and radii of bigger semi-circle is 2 m.



Area of shaded portion A

$=$ Area of semi-circle of radii 3 m $-$ Area of semi-circle of radii 2 m $+ \text{Area of semi-circle of radii 1 m}$

$$= \frac{1}{2} \pi (3)^2 - \frac{1}{2} \pi (2)^2 + \frac{1}{2} \pi (1)^2$$

$$= \frac{1}{2} \pi (9 - 4 + 1) = 3\pi$$

Area of portion B

$= 2$ [Area of semi-circle of radii 2 m $-$ Area of semi-circle of radii 1 m]

$$= 2 \left[\frac{1}{2} \pi (2)^2 - \frac{1}{2} \pi (1)^2 \right] = 3\pi$$

Similarly, area of shaded portion C

$=$ Area of portion A

Hence, the ratio of areas A , B and C is $1 : 1 : 1$.